Fiscal Year:	FY 2026	**Task Last Updated:**	FY 11/12/2025
PI Name:	Dunand, David Ph.D.		
Project Title:	Microstructure Evolution in Freeze-Cast Materials		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60208	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	11/06/2017	**End Date:**	03/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Strutzenberg Littles, Louise	**Contact Phone:**	(256) 544-0946
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Flight Program:	ISS		
Flight Assignment:	Note: End date changed to 03/31/2025 per NSSC information (Ed., 12/3/24). Note: End date changed to 11/05/2024 per NSSC information (Ed., 3/4/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Voorhees, Peter Ph.D. (Northwestern University)		
Grant/Contract No.:	80NSSC18K0196		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Freeze-casting is a novel technique that utilizes ice as a fugitive space holder to fabricate a diverse variety of materials exhibiting elongated, aligned pores. Experimental studies in microgravity inherently simplify the freeze-casting system by minimizing gravity-induced forces that contribute to its complexity, e.g.. sedimentation, buoyancy, and natural convection. Freeze-casting has the potential to produce porous products with specific microstructure including net- and complex-shaped products, provided solidification conditions are properly controlled. Moreover, freeze-casting holds significant promise as an in situ resource utilization technique for space-based materials processing, thus increasing the reliability and safety of access to space while also decreasing overall costs. An improvement in scientific knowledge entails robust and predictive control of materials for a wide variety of applications, thus enabling optimized fabrication on Earth, on planetary surfaces (Moon and Mars), and in orbit.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Freeze-casting has the potential to produce porous products with specific microstructure including net- and complex-shaped products, provided solidification conditions are properly controlled. Moreover, freeze-casting holds significant promise as an in situ resource utilization technique for space-based materials processing, thus increasing the reliability and safety of access to space while also decreasing overall costs. An improvement in scientific knowledge entails robust and predictive control of materials for a wide variety of applications, thus enabling optimized fabrication on Earth, on planetary surfaces (Moon and Mars), and in orbit.
Task Progress:	Per the Principal Investigator (Ed., 11/12/25): FILLING OF AMPOULE #4 AND #5 Two ampoules were filled with suspensions: (i) naphthalene and Aerosol-OT (AOT) in Ampoule #4; and (ii) naphthalene, Cu particles (0.5 µm diameter), and AOT in Ampoule #5, with the upgraded filling station. The filling length in the liquid state is about 60 mm. The first step was to connect the customized glass funnel with an assembled ampoule and then evacuate the whole system to pressures of about 180-200 mTorr. After evacuating, two stopcocks between the glass funnel and the ampoule were switched off to maintain the vacuum level in the ampoule. The upper glass funnel was purged with UHP helium and then loaded with scientific materials. The glass funnel was sealed and then evacuated to vacuum levels of 300-330 mTorr. Meanwhile, the thermal batch was filled with water and heated up to ~93 °C to melt loaded materials. When all naphthalene was melted, the magnetic stirrer was switched on to mix scientific materials (4 h for suspension in Ampoule #4 and 7 h for suspension in Ampoule #5) while sonicating the thermal bath. After mixing the suspensions, the stopcocks were turned on to fill the ampoules with a length of about 6 mm and then turned off immediately to prevent over-filling. The thermal bath was drained out hot water immediately and then refilled with ice water to quench filled suspensions and prevent sedimentation during solidification. AMPOULE SEALING After solidification, the filled ampoules were evacuated and sealed with Kovar piston and plug with a customized sealing station provided by Redwire. The ampoule was inserted into a sealing head and then fixed with a hose clamp. The head was then evacuated to a vacuum level of about ~200 mTorr. The Kovar piston was pushed into the ampoule with a steel rod until the upper side of the solidified suspension. A spring and a Kovar plug were then installed to seal the ampoule. Kovar plugs on both sides are glued to ampoule glass with Torr Seal. GROUND TESTING OF AMPOULE #2 AND #3 IN PFMI FURNACE The filled ampoule #2 containing distilled naphthalene and 1 wt.% AOT was placed in the Pore Formation Mobility Investigation (PFMI) furnace vertically and heated up to 87-90 ºC to melt down naphthalene prior to freeze casting. The cold zone was moved from the cartridge side toward the other end with a defined translation velocity to freeze-cast scientific materials. An ampoule was filled with naphthalene with 1 wt.% AOT was tested in the PFMI furnace with a furnace translation velocity 100 µm/s. To understand the effect of furnace configuration on thermal profiles during freeze casting, the gap between the translating electrode and its neighboring cold-zone was dialed into ~ 1.3 cm. The filled ampoule #3 containing distilled naphthalene, 7.47 g copper particles (1 µm diameter), and 1 wt.% AOT was placed in the PFMI furnace vertically and heated up to 87-90 ºC to melt down the filled suspension prior to freeze casting. The cold zone was moved from the cartridge side toward the other end with a translation velocity of 100 µm/s to freeze-cast scientific materials. However, the melting process was not successful without complete melting back of filled suspensions. Only part of the filling was melted during heating. Moreover, notable sedimentation of Cu particles was observed after testing. The higher density of Cu particles induced significant sedimentation during melting and freeze-casting under gravity. It could generate a considerable impact on heat dissipation and temperature gradient during freeze-casting. This issue needs to be considered during post-test analysis of thermal profile data from PFMI furnace ground testing. Nevertheless, the sedimentation of Cu particles should not be an issue during the flight testing on the International Space Station (ISS).
Bibliography Type:	Description: (Last Updated: 09/30/2024)